

Modeling Seminar Paper

Topic 3

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1 Introduction

Hyperuniform systems are defined by the suppression of density fluctuations over large spatial scales. This behavior is quantified via the structure factor $S(K)$, defined as the Fourier transform of the pair correlation function or equivalently as the normalized variance of density fluctuations in reciprocal space. For a finite number of points $\mathbf{r} = \{r_1, \dots, r_N\} \subset \mathbb{R}^d$ it can be computed as

$$S(k, \mathbf{r}) = \frac{|\widehat{\rho}(k, \mathbf{r})|^2}{N} = \frac{1}{N} \left| \sum_{i=1}^N e^{i k \cdot r_i} \right|^2. \quad (1)$$

We call a point pattern hyperuniform if

$$\lim_{|k| \rightarrow 0} S(k) = 0.$$

Originally studied in the context of disordered materials and quasicrystals, hyperuniform structures appear in physics, biology, and photonic materials due to their unique statistical and optical properties [Tor18].

In this work, we consider the efficient and parametrized generation of hyperuniform point patterns and scalar fields. Such configurations can be produced algorithmically (e.g., through collective coordinate optimization) to display different types of hyperuniformity. To analyze the resulting point patterns, we employ persistent homology, a method from topological data analysis that quantifies the birth and death of topological features across multiple scales. Persistent diagrams provide a compact signature of the underlying structure, offering insight into the degree and nature of hyperuniformity in a way that complements spectral-based metrics.

2 Generating hyperuniform point patterns using the collective coordinate approach

This section is largely based on [Sal+24], Section II. The basic idea of the collective coordinate approach is to define an objective structure factor S_0 to optimize the particle positions towards. Let $\mathbf{r} = \{r_1, \dots, r_N\} \subset \mathbb{R}^d$ be particle positions and $\mathcal{K}$ a set of wave vectors in reciprocal space. We define a loss like

$$\mathcal{L}(\mathbf{r}) = \sum_{k \in \mathcal{K}} w(k) |S(k, \mathbf{r}) - S_0(k)|^2$$

where w is a weighting function. For our purposes we define

$$S_0(k) = \begin{cases} K^{-\alpha}(1 - H)|k|^\alpha + H, & |k| < K \\ 1, & \text{else} \end{cases}$$

and let $w(k) = |k|^{-(d-1)}$ so constraints are equally strong on every spherical shell [SCM24], as S_0 exhibits radial symmetry. The parameter α controls the

general shape of the map in a power-law fashion while H controls the degree of hyperuniformity, where H should be small such that $S_0(k) \approx 0, |k| \rightarrow 0$. Finally the parameter K controls the area in reciprocal space where we constrain fluctuations. (Note that for a poisson process the structure factor is 1 [Tor18], therefore constraining the structure factor to 1 leads to effectively random density fluctuations on small scales.)

In order to use a fast optimizer we can compute the derivative of the loss

$$\begin{aligned}
\frac{d\mathcal{L}}{dr_i} &= \sum_{k \in \mathcal{K}} 2w(k)(S(k, \mathbf{r}) - S_0(k)) \frac{dS}{dr_i} \\
&= \sum_{k \in \mathcal{K}} \frac{2w(k)}{N} (S(k, \mathbf{r}) - S_0(k)) (\hat{\rho} \frac{d\bar{\rho}}{dr_i} + \bar{\rho} \frac{d\hat{\rho}}{dr_i}) \\
&= \sum_{k \in \mathcal{K}} \frac{2w(k)}{N} (S(k, \mathbf{r}) - S_0(k)) (-\hat{\rho} \mathbf{i} k e^{-\mathbf{i}k \cdot \mathbf{r}_i} + \bar{\rho} \mathbf{i} k e^{\mathbf{i}k \cdot \mathbf{r}_i}) \\
&= \sum_{k \in \mathcal{K}} \frac{-2kw(k)}{N} (S(k, \mathbf{r}) - S_0(k)) (\hat{\rho} \mathbf{i} e^{-\mathbf{i}k \cdot \mathbf{r}_i} + \overline{\hat{\rho} \mathbf{i} e^{-\mathbf{i}k \cdot \mathbf{r}_i}}) \\
&= \sum_{k \in \mathcal{K}} \frac{-2kw(k)}{N} (S(k, \mathbf{r}) - S_0(k)) (2\text{Re}(\hat{\rho} \mathbf{i} e^{-\mathbf{i}k \cdot \mathbf{r}_i})) \\
&= \text{Re}(\sum_{k \in \mathcal{K}} \frac{-4\mathbf{i}kw(k)}{N} (S(k, \mathbf{r}) - S_0(k)) \hat{\rho} e^{-\mathbf{i}k \cdot \mathbf{r}_i}) \tag{2}
\end{aligned}$$

For the wave vectors we notice that $S(k) = S(-k)$ and $S_0(k) = S_0(-k)$ and therefore we only choose wave vectors on a half plane. Then we have $M := |\mathcal{K}|$ independent constraints. Since we have N points in d dimensions the total number of degrees of freedom is simply dN and therefore $M < 2dN$ should be given, where the factor 2 comes again from the symmetry, so that we don't over-constrain the points positions. Generally $\frac{M}{2dN} \leq 0.5$ is used [SCM24].

2.1 Optimization using nuFFT

In [SCM24] an more efficient algorithm is proposed which utilizes non-uniform fast Fourier transforms (nuFFT) to speed up the computation of the structure factor and derivative in particular. We can use nuFFT to evaluate for $x_1, \dots, x_I \in \mathbb{R}^d$, $c_1, \dots, c_I \in \mathbb{C}$ and $y_1, \dots, y_J \in \mathbb{R}^d$ an expression of the form

$$f_j = \sum_{i=1}^I c_i e^{\pm \mathbf{i} y_j \cdot x_i}, \quad j = 1, \dots, J$$

in $\mathcal{O}(N \log N)$ which is better than the direct approach to evaluating the sums, which takes $\mathcal{O}(N^2)$ (assuming M scales linearly with N) [SCM24]. To achieve efficiency, the nuFFT approximates the nonuniform transform by spreading (or "smearing") each input point onto a nearby uniform grid using smooth kernel

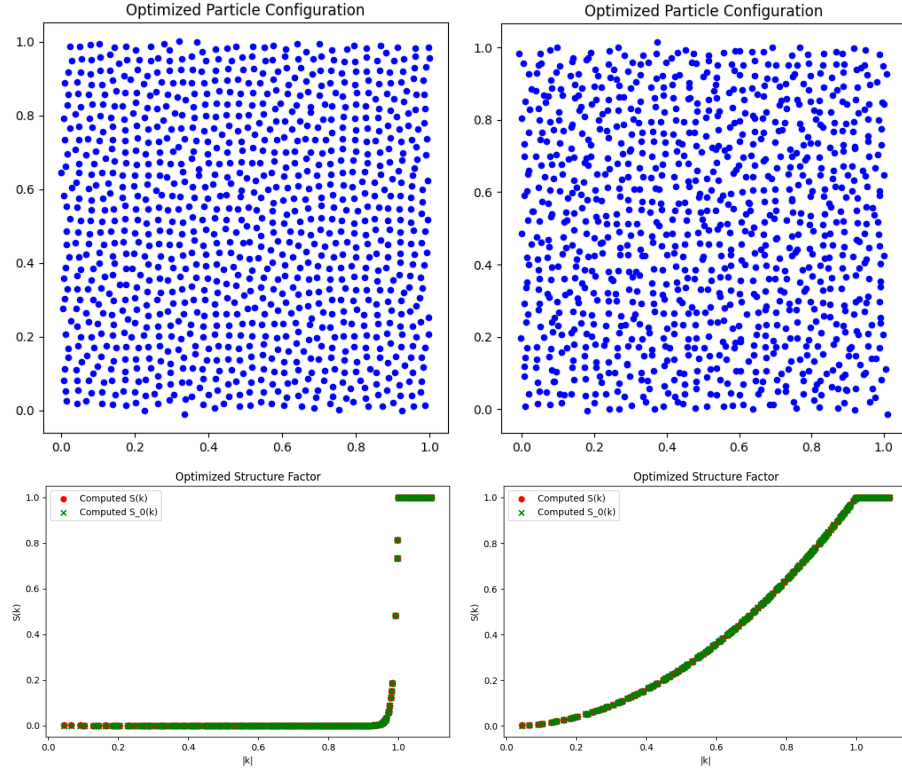


Figure 1: Two hyperuniform point patterns of $N = 1024$ (top) and their structure factors compared to the objective structure factors (bottom) using $K = 44\pi$, $H = 10^{-4}$ and $\alpha = 100$ (left) and $\alpha = 2$ (right).

functions. A standard FFT is then applied on the grid, and the result is interpolated back to the desired nonuniform output frequencies via convolution with the same kernel. Approximation errors arise from the gridding and interpolation step. The accuracy depends on the tolerance and oversampling factor, which can be tuned to balance speed and precision [FIN24].

Equations (1) and (2) for the structure factor and the gradient of the loss function exhibit such a form and may thus be computed using nuFFTs with

$$c_i = 1$$

for $i = 1, \dots, N$ and

$$c_k = \frac{-4ikw(k)}{N}(S(k, \mathbf{r}) - S_0(k))\hat{\rho}(k, \mathbf{r})$$

for $k \in \mathcal{K}$ respectively. We used fiNUFFT ("fast iterative nuFFT") [FIN24] for an implementation with a python interface. We benchmarked the naive

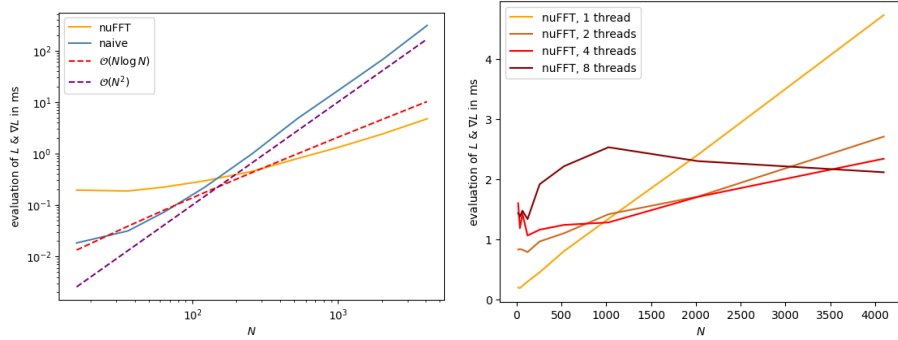


Figure 2: Benchmark of the objective function and gradient function for different N with $M \approx 0.4 \cdot 2dN$ comparing the naive approach and the nuFFT acceleration (left) and nuFFTs scaling for different number of threads (right).

method and the nuFFT accelerated method for different number of points N . As expected, the nuFFT accelerated method exhibits $\mathcal{O}(N \log N)$ complexity while the naive approach scales like $\mathcal{O}(N^2)$. For relatively small $N \leq 10^2$ however, the naive approach is faster.

For one full generation of a point pattern we summarize the results in table 1. Importantly we see that the minimizer has to evaluate the function and gradient in the naive method more often than when using nuFFT. As nuFFT uses some interpolation, there could be some smoothing which makes it easier for the minimizer to run. We checked that the final loss value for the loss using nuFFT is the same than the loss on the point pattern using the naive approach. We conclude that the nuFFT method is indeed practical to use. It's potential is further illustrated in figure 2 (left), as we can multithread nuFFT easily which can improve performance for large N significantly.

	naive	nuFFT
Function evaluations	1519	1004
Gradient evaluations	1519	1004
Iterations	1000	1000
Loss value	$\approx 2.6 \cdot 10^{-4}$	$\approx 3.0 \cdot 10^{-4}$
Time	11m 24.2s	6m 47.6s

Table 1: Result of running SciPy’s minimize (BFGS) for 1000 iterations and $N = 1024$.

We note here that although care has been taken to efficiently implement the method, further optimization is possible. For example by using a custom minimizer which allows reusing already computed values.

3 Random scalar fields and hyperuniformity

Recently, the idea of hyperuniformity has been extended to random scalar fields. We concentrated on expanding our knowledge of the hyperuniformity of scalar fields since the study of random scalar fields is simpler. Numerous physical contexts can give rise to scalar fields, such as laser speckle patterns, temperature and concentration fields in porous and heterogeneous media, and spatial patterns in biological, chemical, and ecological systems. It has previously been demonstrated that the hyperuniformity of the original point configurations will be inherited by scalar fields created by encompassing hyperuniform point configurations with a non-negative dimensionless radial scalar kernel function.

The auto-covariance function is defined as follows for a statistically homogeneous random scalar field $F(\mathbf{x})$ in $\mathbb{R}^d$ that has a real value:

$$\psi(\mathbf{r}) = \langle (\mathbf{F}(\mathbf{x}_1) - \langle \mathbf{F}(\mathbf{x}_1) \rangle)(\mathbf{F}(\mathbf{x}_2) - \langle \mathbf{F}(\mathbf{x}_2) \rangle) \rangle \quad (3)$$

The spectral density function $\tilde{\psi}(\mathbf{k})$ is just the Fourier transform of the auto-covariance function $\psi(\mathbf{r})$ in the case $\mathbf{r} = \mathbf{x}_2 - \mathbf{x}_1$.

The hyperuniformity condition is given by:

$$\lim_{|\mathbf{k}| \rightarrow 0} \tilde{\psi}(\mathbf{k}) \quad (4)$$

Implying the sum rule:

$$\int_{\mathbb{R}^d} \psi(\mathbf{r}) \, d\mathbf{x} = 0 \quad (5)$$

The volume integral of $\psi(\mathbf{r})$, which corresponds to a homogeneous hyperuniform scalar field, must vanish identically due to the presence of both positive and negative correlations.

3.1 Generating hyperuniform gaussian fields

Based on [MT17], Section IV, A.

A superposition of plane waves with random wave vectors and phases creates a Gaussian random field, $f(\mathbf{r})$:

$$f(\mathbf{r}) = \frac{1}{\sqrt{N}} \sum_{i=1}^N \mathbf{A}_i \cos(\mathbf{k}_i \cdot \mathbf{r} + \phi_i) \quad (6)$$

$\mathbf{k}_i$ are the directions, directed uniformly and randomly. ϕ_i the phase, are random numbers lying in $[0, 2\pi]$.

In order to achieve unity auto-covariance at the origin, we normalise the field: $\psi(\mathbf{r} = \mathbf{0}) = \mathbf{1}$. Then we get that A_i satisfying:

$$\frac{1}{2N} \sum A_i^2 = 1 \quad (7)$$

For the spectral density $\tilde{\psi}(\mathbf{k})$, we get the following in the continuous limit:

$$\psi(\mathbf{r} = \mathbf{0}) = \int_{\mathbb{R}^d} \tilde{\psi}(\mathbf{k}) = \mathbf{1} \quad (8)$$

Any higher-order correlation function can be fully defined by the two-point information shown in $\psi(\mathbf{r})$ through the correlation matrix for Gaussian variables with zero mean.

Wave vector components $\mathbf{k}_i$ that meet a particular auto-covariance criteria can be added together to create these Gaussian random fields. One can select amplitude functions $A(\mathbf{k})$ that vanish at $\mathbf{k} = \mathbf{0}$ to produce hyperuniform scalar fields, making sure the spectral density likewise disappears at that point. Resulting because the spectral density depends on $A(\mathbf{k})^2$ and simply requires a convergent volume integral.

Accurate approximations are made possible by increasing resolution, even though numerical implementations are restricted to discrete $\mathbf{k}$ -space and finite systems.

The general formula is:

$$f(\mathbf{r}) = \mathbf{C} \sum_{i=1}^N \mathbf{A}(\mathbf{k}_i) \cos(\mathbf{k}_i \cdot \mathbf{r} + \phi_i) \Delta(\mathbf{k}_i^d), \quad (9)$$

$\mathbf{C}$ is the normalisation constant, and $\Delta(k_i^d)$ is the volume element in d -dimensions containing $\mathbf{k}_i$.

For the amplitude function we applied two choices in 2-D:

$$A(k) = \sqrt{k} e^{\alpha k^2} \quad (10)$$

$$A(k) = \frac{k^2}{\sqrt{\beta + k^{10}}} \quad (11)$$

With both α and β as positive constants. In both situations, the volume integral of $A^2(k)$ converges and $A(k)$ disappears at the origin.

We can achieve a stealthy version of our formula by modifying the formula with an implementation of a translation onto the spectral density. That causes us to get:

$$\tilde{\psi}_{stealthy}(k) = \begin{cases} \tilde{\psi}_{hyperuniform}(k - K), & k \geq K; \\ 0, & k < K. \end{cases}$$

Therefore, applying this to (3.1) yields:

$$f(\mathbf{r}) = \mathbf{C} \sum_{i=1}^N \mathbf{A}(\mathbf{k}_i) \cos((\mathbf{k}_i + K\hat{\mathbf{k}}_i) \cdot \mathbf{r} + \phi_i) \Delta(\mathbf{k}_i^d), \quad (12)$$

where K is chosen to be $10\Delta k$.

The hyperuniformity metric here is relative small and given by:

$$H \equiv \frac{\tilde{\psi}(0^+)}{\tilde{\psi}(k_{peak})} \quad (13)$$

4 Persistent homology analysis

Based on [Sal+24], Section III.

Persistent homology is a method from topological data analysis (TDA) that studies the multi-scale topological features of a space. It tracks the birth and death of homological features such as connected components, holes, and voids across different scales. (...)

4.1 Experiment on S_0 with a bump

In nature, structure factors of hyperuniform systems often display a bump. We can multiply S_0 from section 2 with a simple gaussian of the form

$$G(x_K) = 1 + A \cdot \exp\left(-\frac{(x_K - x_0)^2}{2\sigma^2}\right), \quad x_K(x) := \frac{x}{K}$$

with K the cutoff parameter and x_0, σ, A typical parameters for the gaussian.

When generating point patterns for structure factors with a bump we can not visually distinguish a configuration coming from an imposed structure factor

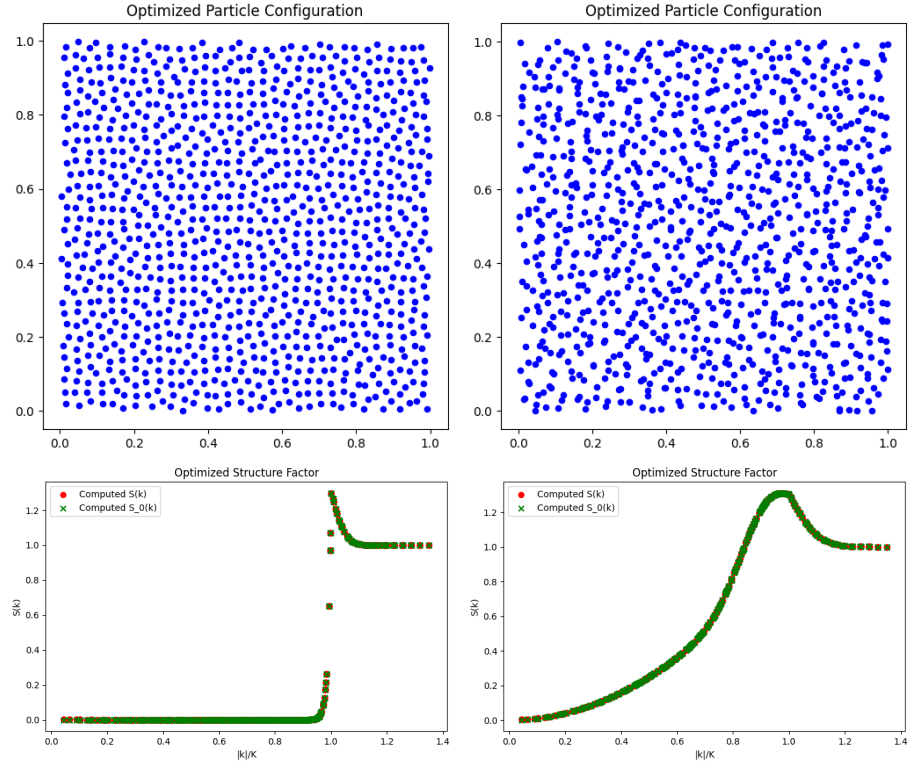


Figure 3: Two hyperuniform point patterns of $N = 1024$ (top) and their structure factors compared to the objective structure factors (bottom) with bumps using $A = 1.5, x_0 = 0.95, \sigma = 0.5$ (left) and $A = 0.5, x_0 = 0.9, \sigma = 0.5$ (right) and α, K, H as in figure 1.

with or without a bump (Compare figure 1 and figure 3). We use the persistence diagrams to explore whether such bumps are quantifiable in the topology of the point patterns. To this end, we generated 10 point patterns like 3 (left) with and without a bump each for different perturbed initial point configurations. We then compute the persistence diagram and the pairwise wasserstein distances between the point patterns without the bump to each other and to the point patterns with a bump. Finally, we compute mean and variance $[\mathbb{E}; \mathbb{V}]$ of these distances. As one would expect the mean of comparing bump and no bump con-

	no bump vs. no bump	no bump vs. bump
h_0	$[0.192; 1.1 \cdot 10^{-3}]$	$[0.264; 3.3 \cdot 10^{-3}]$
h_1	$[0.231; 6.0 \cdot 10^{-4}]$	$[0.258; 6.3 \cdot 10^{-4}]$

Table 2: Mean and variance of wasserstein distances of h_0 and h_1 from persistence diagrams.

figurations is larger than comparing no bump configurations to each other. The largest difference is observed in the h_0 component of the persistence diagram, reflecting shifts in connected component structure. Although the difference is still relatively small, it demonstrates how the illusiveness of hyperuniformity can effectively be analyzed using the persistent homology.

5 Conclusion

In this work, we generated a range of parameterized hyperuniform point patterns and gaussian scalar fields using multiple generation techniques. By benchmarking these methods, we evaluated their flexibility and effectiveness in producing controlled spatial structures with tunable spectral properties. We applied persistent homology as a topological tool to analyze and compare the underlying structure of these patterns, revealing subtle differences not apparent through visual inspection alone.

The versatility of our generation methods allows for the creation of not only disordered hyperuniform structures but also patterns with features reminiscent of quasiperiodicity, depending on parameter choices. This opens pathways for connecting our approach to related topics within the seminar, such as models of quasicrystals and suggests persistent homology as a promising framework for structural classification across these domains.

Literature

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Appendix

The code and documented results can be found on GitHub.

Statement of Contributions

Enno Igel: Sections 2, 2.1, 4.1 (+ 1, 5)
Bekzat Zhardembayev: 4
Abdul Salam Farooqi: Section 3